

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
 II SEMESTER 2013-2014
 EEE/CS/INSTR F241 MICROPROCESSOR PROGRAMMING AND INTERFACING
 COMPREHENSIVE EXAMINATION-A (OPEN BOOK)
 13-05-2014

MARKS: 30

ID No:	Name:	DURATION: 60 MIN
Sec:		

Note: This in-built question/answer paper contains fifteen questions, each carrying 2 marks. Use last few pages of answer sheet of PART-B to do your rough work. You can submit PART-A anytime and collect PART-B question paper.

Recheck Request:

Q1. The expression $No. \text{ of clocks} = 22(CX-1) + 1,441,802 (DX-1) + 36$ introduces a delay of 0.758 sec for CX=4150H and DX=0006H in an x86 CPU based program. The frequency of operation of CPU is 10 MHz. Compute the minimum and maximum delay which could possibly be achieved using the expression.

Q2. A MACRO is written by a user to push the contents of EAX, EBX, ECX and EDX registers in order onto the stack. Show the contents using register names and their physical addresses in the stack, after the code associated with the MACRO is executed. Initially, assume SS=0000H and SP=1000H.

Q3. If a media has 256 bytes/sector and the maximum number of root directory entries specified in BIOS Parameter Block is 612, then total number of sectors reserved for root directory is:

Q4. If the 8 byte descriptor of a segment in 80286 is 00 00 FF 32 00 00 00 FF, then the size and the starting address of the segment is

Q5. If AX=FFFFH and CL = 02H. What is the result of executing the instruction DIV CL?

Q13. The Intel 8088 consists of a bus interface unit (BIU) and an execution unit (EU), which forms a 2-stage pipeline. The BIU fetches instructions into a 4-byte instruction queue.

a) Suppose the tasks performed by the BIU and EU take about equal time. By what factor does pipelining improve the performance of the 8088? Ignore the effect of branch instructions.

b) Repeat the calculation assuming that the EU takes twice as long as the BIU.

Q14. Consider a system employing interrupt-driven I/O for a particular device that transfers data at an average of 8 KB/s on a continuous basis. Assume that interrupt processing takes about 100 μ s (i.e., the time to jump to the interrupt service routine (ISR), execute it, and return to the main program). Determine what fraction of processor time is consumed by this I/O device if it interrupts for every byte.

Q15. A pipelined processor has a clock rate of 2.5 GHz and executes a program with 1.5 million instructions. The pipeline has five stages, and instructions are issued at a rate of one per clock cycle. What is the throughput (in MIPS) of the pipelined processor?

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EEE/CS/INSTR F241 MICROPROCESSOR PROGRAMMING AND INTERFACING
COMPREHENSIVE EXAMINATION-B (OPEN BOOK)

13-05-2014

DURATION: 120 MIN

MARKS: 60

Note: The paper contains four questions.

Q1. An 8086 system needs memory to be interfaced as per following specifications.

(15)

2KB SRAM from address F4000H to F4FFFH and from address F8000H to F8FFFH
2KB ROM from address 02000H to 03FFFH and from address F0000H to F1FFFH

The following hardware components are available:

- 2KB SRAM chips. (4 nos.), 4KB ROM chips (4 nos.)
- 74LS138 (01 no.)
- Two input OR gates (10 nos.), Four input OR gate (03 nos.)
- Two input AND gate (03 nos.), Three input AND gate (03 nos.)
- Inverters (07 nos.)

Q2. A 256 byte RAM chip and ADC 570 is to be interfaced with 8086 using 8255. The base address for 8255 is 40H. The ADC 570 accepts an input signal ranging between -50mV to +50mV from a sensor and sends out the digital data to the 8255 which in turn writes it to the top location of the memory. The control signals necessary to start and detect end of conversion are provided by 8255. Address, Data and Control lines of memory are to be connected through ports of 8255. Give the complete interfacing of the system and write a program to store the digitized data into the memory. Assume the data bus of the ADC 570 is already cleared before the conversion takes place. The 8255 is interfaced with lower byte of the data bus of 8086. (15)

Q3. Using 8254 and required gates, generate the waveform as shown below. Interface 8254 at base address 90H and show the decoding logic (fixed address). You are given a 1Hz clock source. Write a procedure called "INIT" which will initialize 8254 to generate the given waveform. (15)



Q4. (a) A disk controller usually has 8-bit data bus output and it uses DMA controller to transfer data bytes to memory. If we connect this 8-bit data bus to either D0-D7 or D8-D15 of the 8086 system, then we would be able to transfer only byte size data to either even or odd addresses of memory. Suggest a hardware interfacing scheme to resolve this problem. (5)

(b) Write a program to initialize 8237 to transfer a data block of size 4KB available at 5000H:0000H to 5000H:1000H. The base address of 8237 is F0H. Initialize the 8237 for normal timings, fixed priority, and extended write with DREQ active high and DACK active high. The 8237 should work in auto initialization mode with address increment, block mode select with read transfer on channel 0. (10)